

Harsh Raj

AI Engineer | Generative AI & LLM Systems

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PROFESSIONAL SUMMARY

AI Engineer and Backend Developer specializing in Generative AI, Large Language Models (LLMs), and production-grade Agentic AI systems. Skilled in building scalable backend services with Python, FastAPI, and SQL databases, and in designing LLM-powered applications using LangChain, LangGraph, RAG, and vector search (FAISS). Proven ability to architect multi-step AI agent workflows, automate complex business processes, and deliver reliable, low-latency APIs from prototype to production. Strong foundation in System Design, Data Structures and Algorithms, Cloud Computing (AWS, GCP), and full-stack development, with a track record of translating ambiguous requirements into measurable engineering outcomes.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL, HTML, CSS

AI / LLM Frameworks: LangChain, LangGraph, RAG, Retrieval-Augmented Generation, AI Agents, Agentic AI, Prompt Engineering, Prompt Chaining, Function Calling, Tool Calling, Context Management, Memory, Embeddings, Semantic Search, FAISS, Hugging Face, Transformers, Gemini API, OpenAI API, NLP, OCR

Backend Technologies: FastAPI, Uvicorn, REST API, Microservices, CRUD, Workflow Automation, State Machines, JSON, API Integration, Inference

Frontend: React.js, Node.js, Streamlit, JavaScript (Fetch API), HTML/CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLAlchemy, Vector Database

Cloud & DevOps: AWS, GCP, Railway, Docker, Git, GitHub, CI/CD, Linux, Production Deployment

Core Computer Science: Data Structures, Algorithms, System Design, Problem Solving, Scalable Systems, Performance Optimization, DBMS, Pandas, NumPy, Power BI

PROFESSIONAL EXPERIENCE

AI Engineer | Croccle May 2025 - May 2026

- Engineered NLP pipelines to process voice-based and conversational search queries, improving query intent recognition accuracy by approximately 20% over baseline keyword matching.
- Designed and tested intent-classification algorithms that increased search relevance and personalization, reducing irrelevant result rates across high-traffic query segments.
- Integrated structured data and schema markup into search infrastructure, improving machine readability for AI-driven search engines and accelerating content indexing.
- Researched and benchmarked AI-powered voice assistants (Google Assistant, Alexa, Siri) to evaluate emerging conversational AI trends and inform digital marketing strategy.
- Delivered data-driven recommendations for voice search optimization, contributing to measurable gains in user engagement and session retention.

PROJECTS

Conversational AutoStream AI Agent | Python, LangGraph, Gemini API, RAG, Vector Search, FAISS, JSON Knowledge Base, python-dotenv [GitHub](#)

- Architected a production-ready, multi-agent conversational AI system for a SaaS platform using LangGraph and Gemini API, orchestrating structured state-machine workflows for reliable, repeatable execution.
- Built an intent-classification engine (Greeting, Inquiry, High-Intent Lead) paired with RAG-based retrieval over a JSON knowledge base, improving response accuracy and reducing hallucinated answers.
- Implemented multi-turn conversation memory and automated lead-capture workflows, cutting manual lead-qualification effort by an estimated 30% and improving conversion-ready engagement.
- Engineered secure API key management and modular tool-calling architecture, enabling reliable request routing, fault tolerance, and horizontal scalability for production deployment.

Agentic NLP Chatbot with CRUD Support | FastAPI, Uvicorn, PostgreSQL, SQLAlchemy, LangChain, LangGraph, JavaScript (Fetch API), HTML/CS [GitHub](#)

- Built a full-stack AI-driven conversational chatbot with FastAPI and LangGraph, exposing NLP-based CRUD operations over a PostgreSQL backend through secure REST API endpoints.
- Designed state-managed, multi-step conversation workflows using LangGraph state machines and LangChain memory, with robust input validation and error handling to minimize failed transactions.
- Engineered end-to-end database integration with SQLAlchemy ORM and pycopg2, automating create, read, update, and delete operations through autonomous AI agents and tool execution.
- Optimized request handling and query execution with Uvicorn, improving API response times and supporting concurrent multi-user sessions under load.

EDUCATION

Indian Institute of Information Technology, Pune (IIIT Pune) Pune, India
Bachelor of Technology in Computer Science and Engineering Aug 2022 - May 2026

ACHIEVEMENTS

Ranked in the top 1.5% of Joint Entrance Examination (JEE) 2022, among over 1 million candidates nationwide.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Led Horizon (IIIT Pune) Photography Club as Core Head, overseeing club operations and coordinating a team of contributors.
- Directed public relations for IConclave '23 (IIIT Pune), managing communications strategy for the Annual Techno-Cultural Event.
- Served as Lead Designer and Photographer for Eminence 2024, the official college magazine of IIIT Pune.